

Label Quality

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can write an observable label rule with boundaries and a review option.

I can measure agreement using matched items and an explicit denominator.

I can investigate disagreement and record justified label decisions.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Ambiguous definitions and inconsistent annotation place an upper bound on learnable performance.



LEARNING TARGET Two labelers agree on 16 of 20 leaves. Both used a rule that mistakenly calls shadows 'spots'. Compute agreement and explain why it does not establish correct labels.

Builder name and experiment date:

Readiness for the connected explanation

Why should an unknown duration remain different from zero in a dataset card?

This week's end goal

By the end, I can explain label quality using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use label quality to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

A model learns from examples with recorded targets. If two people interpret a target differently, their labels can mix different tasks. This week you will write a label rule, compare independent labels and keep a record of disputed cases. We use invented corridor descriptions, not real images or a system for controlling a robot.

The classroom task is to label whether a marked practice lane is blocked in a supplied description. Rule v1 says blocked means an object touches or crosses either lane boundary or lies inside the lane; clear means all objects are outside both boundaries and the entire lane is visible. Use review if visibility or position is insufficient to decide. Review is a workflow status for a person to inspect, not a third physical condition of the lane.

Description · Rule v1 result · Reason

Description: A box lies inside a fully visible lane

Rule v1 result: blocked

Reason: Inside satisfies the rule

Description: A wheel touches the boundary line

Rule v1 result: blocked

Reason: Touching is explicitly included

Description: All objects are outside and the lane is fully visible

Rule v1 result: clear

Reason: Both clear conditions hold

Description: A curtain hides part of the lane

Rule v1 result: review

Reason: Hidden space cannot be checked

These are exercise definitions, not real robot safety rules. Do not infer that a lane is physically safe from this label. The label describes only the supplied evidence under this version of the rule.

3. See how it works

ML MECHANISM

Input → **Model** → **Evidence**

Trace the information, fitted behavior, and evaluation proof separately.



IN

INPUT

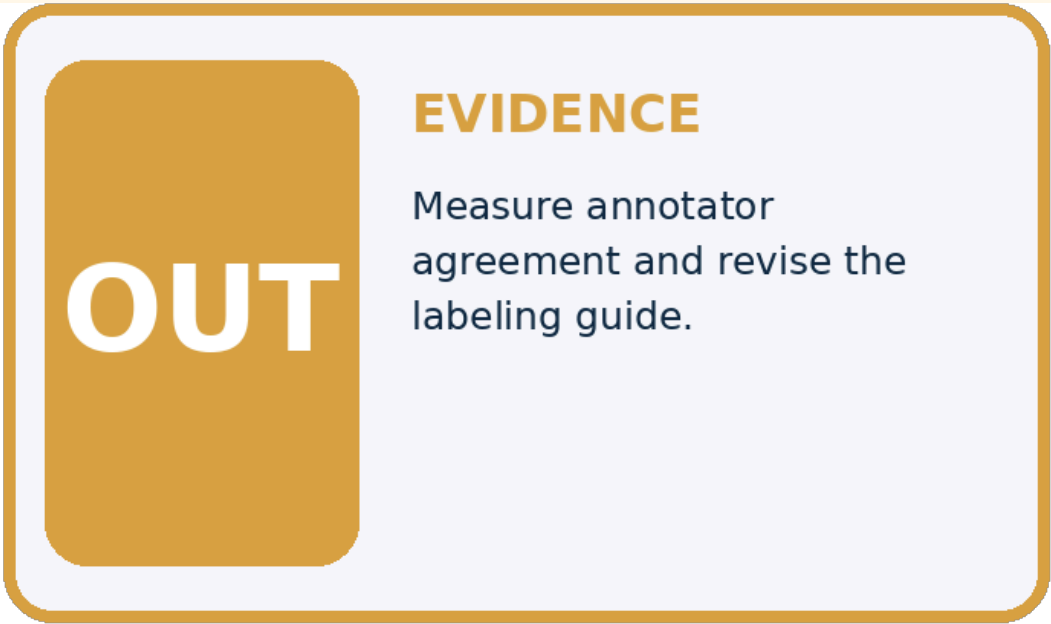
Measure annotator agreement and revise the labeling guide.



ML

MODEL

Ambiguous definitions and inconsistent annotation place an upper bound on learnable performance.



OUT

EVIDENCE

Measure annotator agreement and revise the labeling guide.

QUESTION Measure annotator agreement and revise the labeling guide.

MODEL IDEA Ambiguous definitions and inconsistent annotation place an upper bound on learnable performance.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Two annotators label 20 leaves and agree on 16. Observed agreement = $16 \div 20 = 80\%$.

Draw a fresh data → model → prediction → evaluation trace:

A closer explanation

Two annotators used an earlier vague instruction, mark an obstructed lane, on four invented items. Their labels are shown in the code. A and D match; B and C differ. Compare the same item IDs, not merely the first row in two differently ordered files.

```
pairs = [("A", "blocked", "blocked"),
         ("B", "clear", "blocked"),
         ("C", "review", "clear"),
         ("D", "clear", "clear")]

agree = 0

for item_id, first, second in pairs:
    if first == second:
        agree = agree + 1

print(agree, len(pairs))
```

The output is 2 4: two matching labels among four paired items, so observed exact agreement is 2/4 or 50 percent. Here review is included as a label value in the comparison: review versus clear is a disagreement. This simple count is not model accuracy and does not establish which labels are correct.

Follow a disagreement back to evidence

Observe · Investigate · Record

Observe: B has two labels

Investigate: Read boundary rule

Record: Keep labels and reason

Observe: C includes review

Investigate: Check visible evidence

Record: Retain uncertainty if needed

Observe: A and D match

Investigate: Check rule as well

Record: Agreement is not proof

Preserve independent labels before discussion. Shared mistakes can produce agreement, and missing evidence can prevent a justified final label.

The code assumes one valid pair per unique item ID and allowed label strings. It does not validate IDs or spelling. Use a nonempty paired set before calculating a fraction; zero paired items gives no agreement estimate.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

EVALUATION DESIGN

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL CASE

Expected data from the intended setting



BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



SHIFTED CASE

New subgroup, device, condition, or time

NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: Z receives a new fully visible image with no brown spots.

Prediction: Label the new observation healthy under the toy rule.

Reason: The missing visual evidence is now supplied; the earlier uncertain image was not proof of health.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.



2

LABEL RULE

observable criterion for
assigning a class

3

AGREEMENT

fraction of labels that match

4

ADJUDICATION

resolve disagreements using evidence

Term · Precise meaning · My ML example

Term: annotation

Precise meaning: human assignment of a label

My ML example: _____

Term: label rule

Precise meaning: observable criterion for assigning a class

My ML example: _____

Term: agreement

Precise meaning: fraction of labels that match

My ML example: _____

Term: adjudication

Precise meaning: resolve disagreements using evidence

My ML example: _____

MISCONCEPTION TO REPAIR The model can learn the correct concept even when people label the same example differently.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: Label rule

Meaning: A written definition for assigning an outcome to an example

Word: Annotator

Meaning: A person who assigns labels using a stated rule

Word: Agreement

Meaning: How often annotators assign the same label to the same items

Word: Adjudication

Meaning: Reviewing evidence and rules to resolve a disputed label or retain uncertainty

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 6 • WORKED EVIDENCE

- 1 Two annotators label 20 leaves and agree on 16.
- 2 Observed agreement = $16 \div 20 = 80\%$.
- 3 Most disagreements concern tiny spots near the boundary.
- 4 Define a minimum spot rule, relabel a fixed subset, and report the new agreement.

STEP 1 Two annotators label 20 leaves and agree on 16.

STEP 2 Observed agreement = $16 \div 20 = 80\%$.

STEP 3 Most disagreements concern tiny spots near the boundary.

STEP 4 Define a minimum spot rule, relabel a fixed subset, and report the new agreement.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

For B, the full lane is visible and a wheel touches its boundary. One annotator treated touching as clear; the other treated it as blocked. Rule v1 explicitly includes touching, so the reviewed label is blocked. Preserve both original labels and record the item, rule version, evidence, decision and reviewer. Do not silently overwrite the originals.

For C, a curtain hides part of the lane. One annotator guessed clear; the other requested review. Rule v1 requires full visibility for clear. The justified status remains review until sufficient evidence is available. More votes cannot reveal the hidden area.

Using rule v1, explain the decisions for B and C. If you exclude C from the original comparison, what is agreement among the remaining pairs? Why must you still report the exclusion?

A decision log could say: B, original clear/blocked, wheel touches line, v1, blocked after review by the teacher. For C it could say: original review/clear, lane partly hidden, v1, unresolved review. Add the labeling method and rule version to the dataset card.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Label Quality

Write a label rule that resolves the visible boundary case.

The facts we will use

Toy spotted-leaf rule: at least one fully visible brown spot means spotted; occluded leaves require review. X has one visible spot; Y has none; Z is half hidden.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Label Quality

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Apply visible rule

X is spotted; Y is healthy under this exercise's definition.



2. Handle uncertainty

Z requires review; hidden tissue cannot be treated as visibly spot-free.



3. Review disagreement

If an annotator labels X healthy, compare the image and declared spot rule before changing the label.

CHANGE ONE THING

Z receives a new fully visible image with no brown spots.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Apply visible rule

What happens and why: X is spotted; Y is healthy under this exercise's definition.

Step: 2. Handle uncertainty

What happens and why: Z requires review; hidden tissue cannot be treated as visibly spot-free.

Step: 3. Review disagreement

What happens and why: If an annotator labels X healthy, compare the image and declared spot rule before changing the label.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. If an annotator labels X healthy, compare the image and declared spot rule before changing the label. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Visible brown spots are a simplified classroom target, not a diagnosis of plant disease.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR

Tempting claim: The model can learn the correct concept even when people label the same example differently. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE The model can learn the correct concept even when people label the same example differently.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

Report the paired-item count, agreement count, handling of review and the rule version. For the original four-item comparison, say two of four pairs match, including review statuses. If you report a subset, name its selection rule and denominator. This tiny invented set says nothing about reliability on other scenes.

Keep practice for writing label rules separate from final model testing. Resolving labels needs a documented process; reviewing model predictions to choose convenient labels would undermine a fair evaluation. Next week you will separate related examples when designing data splits.

Two annotators agree on every item but both ignore the boundary rule. What does perfect agreement establish, and what does it fail to establish? Give one evidence check to perform next.

8. Practise together

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

A useful rule names the object of judgment, allowed labels, observable evidence, boundaries and an uncertainty route. Include a typical example and a boundary example. Pilot the rule independently on a few new items, compare labels, discuss causes, revise the wording and label new items again. Record which

rule produced each set of labels.

Do not change the rule simply to improve a score or make a model look correct. A rule revision can change the target itself. If touching is changed from blocked to clear, document the changed task, review all affected examples consistently and keep versions separate. Consult task evidence when interpreting a disagreement; a majority vote alone cannot prove truth.

Practise with less help: Interpret agreement

Use workbook W2 for the connected example “Label definitions and disagreement”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Label definitions and disagreement. Proposed explanation: “Agreement is accuracy”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why should two annotators not vote away Z’s missing evidence? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Label Lab: independently label twenty cards, compute agreement, cluster disagreements, revise one rule, and repeat.

1. Write the hypothesis and expected evidence before running.
2. Label Lab: independently label twenty cards, compute agreement, cluster disagreements, revise one rule, and repeat.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

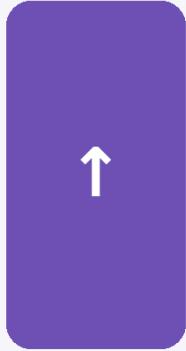
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY

1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: annotation, label rule, agreement, adjudication.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE The model can learn the correct concept even when people label the same example differently.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.



RESPONSIBLE PLANT CLASSIFIER

Write the Plant Classifier label guide with positive, negative, borderline, and not-labelable examples.

THIS WEEK'S CHECKPOINT Write the Plant Classifier label guide with positive, negative, borderline, and not-labelable examples.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

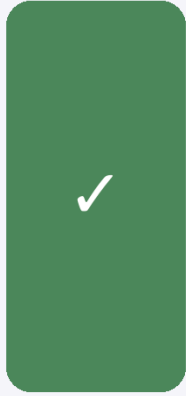
For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF

Measure annotator agreement and revise the labeling guide.

REQUIRED PROOF Measure annotator agreement and revise the labeling guide.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Sampling and Intended Use.

13. Study a complete case

Week 6 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Two annotators disagree on three of ten leaves, especially mild spotting.

The question

Improve label quality.

Worked explanation

Define spotting operationally, inspect disagreement, use appropriate adjudication and document ambiguity. Agreement is 7/10, but agreement is not proof of truth.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Agreement is useful evidence, not truth

Evidence or stage · Value or result

Evidence or stage: Annotator agreement

Value or result: 7/10

Evidence or stage: Disagreement

Value or result: 3/10

Two annotators disagree on three of ten leaves, especially mild spotting.

Improve label quality.

Define spotting operationally, inspect disagreement, use appropriate adjudication and document ambiguity. Agreement is 7/10, but agreement is not proof of truth.

14. Try a new case independently

Independent case: Apply the process independently

New pairs are P blocked/blocked, Q clear/blocked, R review/review, S clear/clear, T blocked/clear. P has an object inside; Q touches the line; R is partly hidden; S has all objects outside a fully visible lane; T has

an object inside. Count agreement including review. Apply v1 to all five. Then report agreement on the original pairs excluding any review status, with its denominator and limitation.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 6 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Two labelers assign different species to the same image. Should you silently pick the label that helps the model score? Describe a better process.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Write the Plant Classifier label guide with positive, negative, borderline, and not-labelable examples.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Two labelers agree on 16 of 20 leaves. Both used a rule that mistakenly calls shadows 'spots'. Compute agreement and explain why it does not establish correct labels.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.